# Karachi Municipal Solid Waste Prediction

## Using Machine Learning Techniques

Data Mining Pipeline Report

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**Course:** Data Mining

**Dataset:** Karachi_Research_Dataset.csv

**Period:** January 2020 – December 2024

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# 1 Introduction

This report outlines the full data mining process used to predict and study Karachi's municipal solid waste (MSW) generation. It covers the key pipeline steps: data loading, preprocessing, exploratory pattern analysis, feature engineering, model building, and result evaluation.

Karachi, Pakistan's largest city, produces millions of tons of solid waste every year. Predicting monthly waste amounts aids city planning, resource allocation, and environmental management. This project draws on **60 months** of data spanning 2020 to 2024, building regression models and identifying seasonal trends through clustering.

The dataset forms a time series on waste output. Factors such as population growth, monsoon rainfall, special events, moisture levels, and waste density shape the target variable. The study compares two regression approaches — *Linear Regression* and *Random Forest* — and applies *K-Means clustering* to reveal distinct waste-generation patterns.

# 2 Dataset Description

The `Karachi_Research_Dataset.csv` file was assembled using municipal records from Karachi's waste management authority, detailing monthly solid waste data from January 2020 to December 2024.

## 2.1 Dataset Overview

**Table 1:** Dataset Overview

| Property | Value |
|---|---|
| Total Records | 60 |
| Features | 11 (original) + 3 (engineered) |
| Target Variable | `Total_MSW_Tons_Month` |
| Date Range | January 2020 – December 2024 |
| Missing Values | None (0 in all columns) |

### 2.2 Key Features

**Table 2:** Key Features in the Dataset

| Feature | Type | Description |
|---|---|---|
| `Total_MSW_Tons_Month` | Target | Monthly waste generation in tons |
| `Population` | Numerical | City population estimate |
| `Plastic_Waste_Tons` | Numerical | Monthly plastic waste (tons) |
| `Generation_Rate_kg_capita_day` | Numerical | Waste generated per person per day |
| `Is_Monsoon` | Binary | 1 = monsoon month, 0 = otherwise |
| `Is_Event` | Binary | 1 = special-event month |
| `Moisture_Content_Percentage` | Numerical | Percentage moisture in waste |
| `Waste_Density_kg_m3` | Numerical | Waste density (kg/m$^3$) |

## 3 Methodology

### 3.1 Data Collection

The `Karachi_Research_Dataset.csv` file was imported using `pandas`, comprising 60 rows and 11 columns. All column headers were verified and the first five rows were examined to confirm data accuracy.

### 3.2 Data Preprocessing

The following preprocessing steps were applied:

- Converted the `Date` column from string to `datetime64` format using `pd.to_datetime()`.
- Extracted `Month` and `Quarter` features to capture seasonality.
- Checked for missing values: **0 missing values** found across all 11 columns.
- Verified data types to ensure all numerical columns were correctly typed for modelling.

### 3.3 Exploratory Data Analysis

Several visualisations were created to understand data distribution and inter-feature relationships (Figures 1–4).

Key EDA findings:

- Total MSW ranged from **255,951 to 382,272 tons/month**.
- Mean generation rate was **0.57 kg/capita/day**.
- Monsoon months (`Is_Monsoon = 1`) occurred in 25% of records.
- Event months (`Is_Event = 1`) accounted for 17% of records.

### 3.4 Feature Engineering

Three new features were derived to improve model performance:

1. **Waste_Per_Capita**: Total waste ÷ Population × 1000 — captures per-person waste intensity.
2. **Plastic_to_Total_Ratio**: Plastic waste ÷ Total waste — measures the plastic fraction.
3. **Is_Summer**: Binary flag set to 1 for months 5, 6, 7 (May–July) — captures summer seasonality.

### 3.5 Model Development

Two regression models were trained using the following 10 features:

`Population, Generation_Rate_kg_capita_day, Is_Monsoon, Is_Event, Moisture_Content_Percentage, Waste_Density_kg_m3, Month, Quarter, Waste_Per_Capita, Is_Summer`

- **Linear Regression:** Features scaled with `StandardScaler`; 5-fold cross-validation applied.
- **Random Forest:** 100 estimators, `random_state=42`; raw features with 5-fold cross-validation.
- **K-Means Clustering:** $k = 3$, applied to four standardised features: `Total_MSW_Tons_Month`, `Plastic_Waste_Tons`, `Moisture_Content_Percentage`, `Waste_Density_kg_m3`.

### 3.6 Evaluation Metrics

- **$R^2$** (Coefficient of Determination) — proportion of variance explained.
- **MAE** (Mean Absolute Error) — average prediction error in tons.
- **RMSE** (Root Mean Squared Error) — penalises larger errors more heavily.
- **Inertia (Elbow Method)** — used to select optimal $K$ for clustering.

## 4 Results and Discussion

### 4.1 Descriptive Statistics

**Table 3:** Descriptive Statistics of Key Variables

| Statistic | Population | Total MSW (Tons) | Plastic Waste (Tons) | Gen. Rate |
|---|---|---|---|---|
| Mean | 17,111,297 | 299,392.34 | 48,367.94 | 0.57 |
| Min | 16,094,000 | 255,950.63 | 40,491.79 | 0.55 |
| Max | 18,169,845 | 382,272.27 | 61,673.80 | 0.71 |
| Std Dev | 614,394 | 27,129.80 | 4,381.78 | 0.04 |

## 4.2 Exploratory Analysis Findings

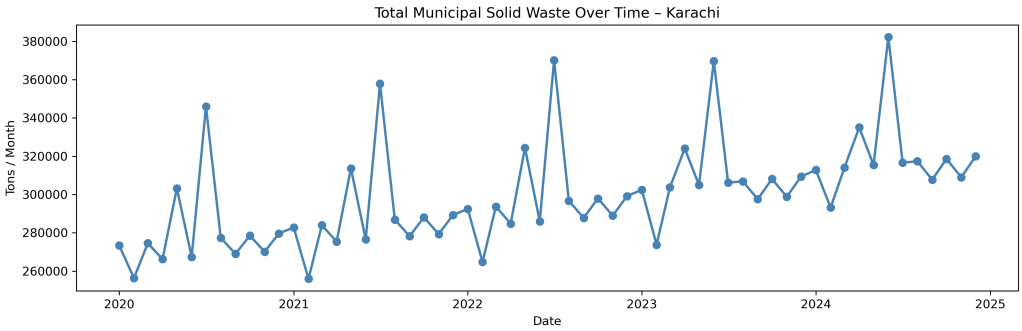



**Figure 1:** Total MSW (Tons/Month) over time (2020–2024). A clear upward trend is visible, rising from approximately 260,000 tons/month in 2020 to over 360,000 tons/month by late 2024, driven by population growth.

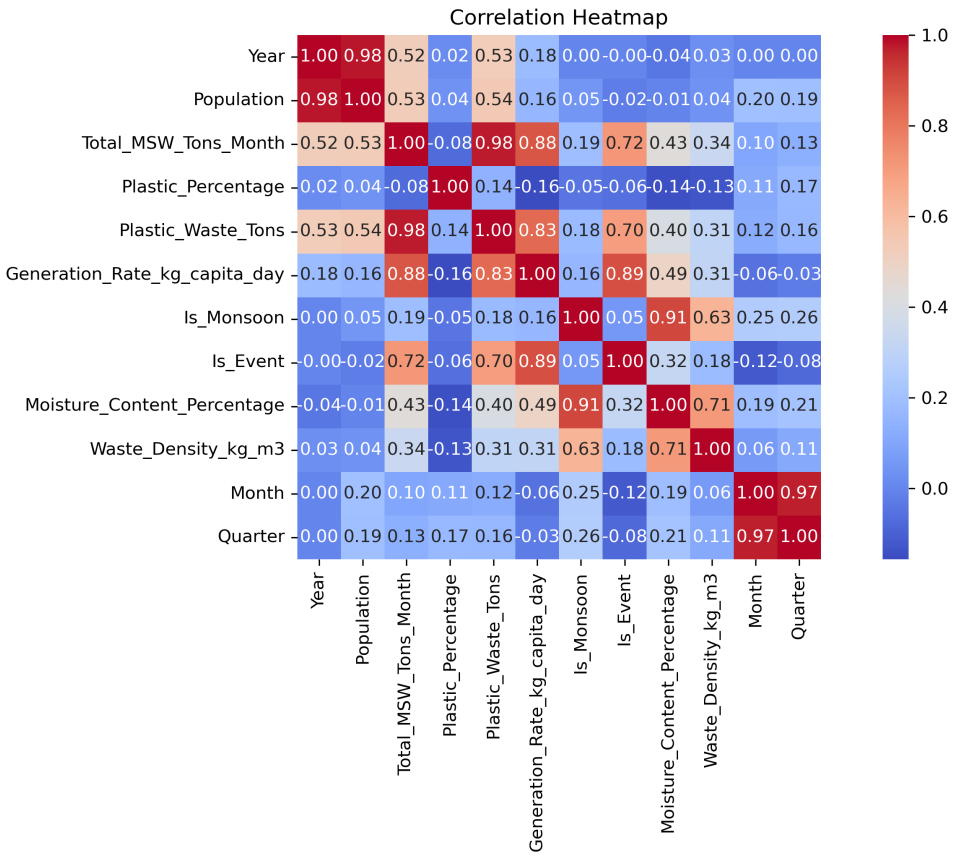



**Figure 2:** Correlation heatmap revealing strong positive correlation between `Population` and `Total_MSW_Tons_Month`, and between `Plastic_Waste_Tons` and total waste.

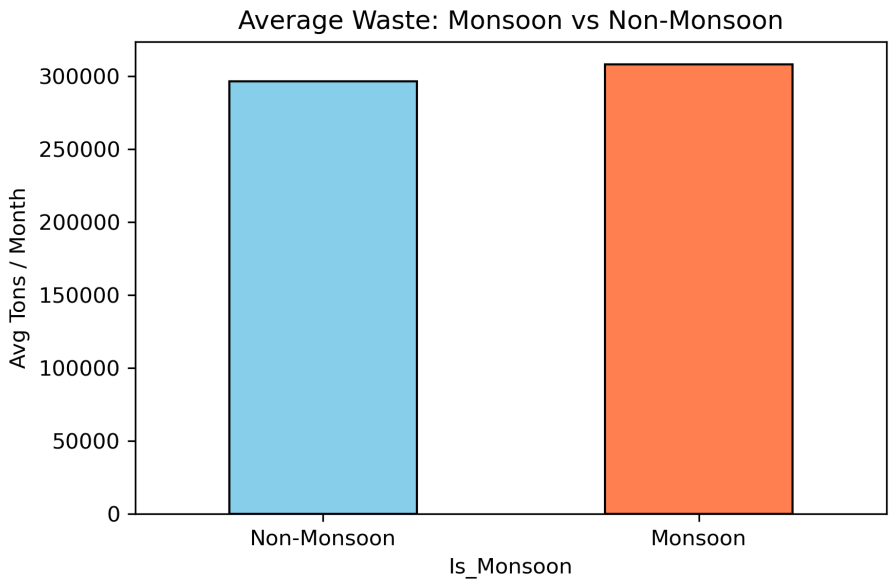



**Figure 3:** Monsoon vs. Non-Monsoon average waste generation. Monsoon months exhibit elevated waste due to increased moisture content and collection challenges.

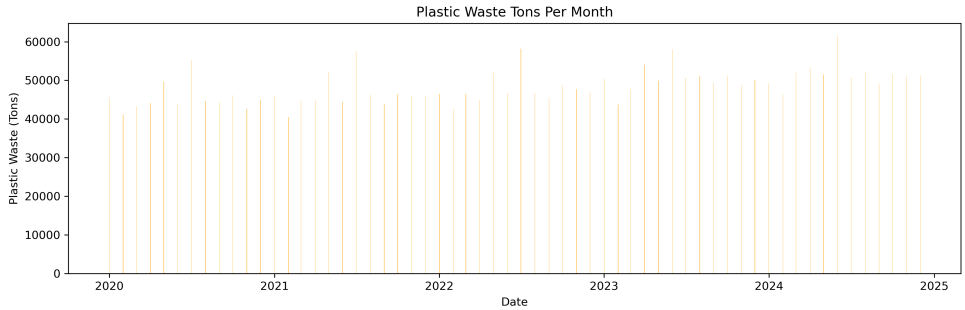



**Figure 4:** Monthly plastic waste (tons) from 2020 to 2024. Plastic waste grew from approximately 41,000 to 60,000 tons/month, tracking closely with overall MSW growth.

## 4.3 Model Performance

**Table 4:** Regression Model Performance Comparison

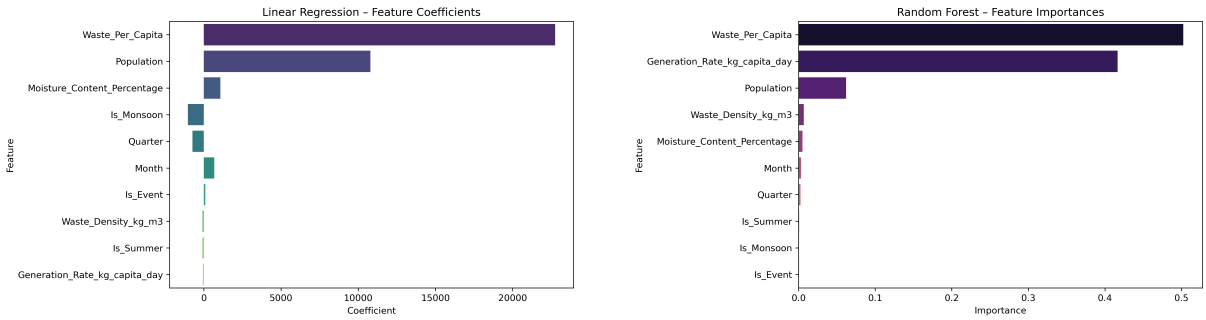
| Model | $R^2$ | MAE (Tons) | RMSE (Tons) |
|---|---|---|---|
| Linear Regression | 0.9984 | 615.16 | 880.55 |
| Random Forest | 0.8413 | 7,656.96 | 8,775.02 |

Linear Regression achieved an outstanding fit with $R^2 = 0.9984$, explaining **99.84%** of variance in monthly waste generation. An MAE of 615 tons against a mean waste of ≈299,000 tons represents an error of less than **0.21%** — excellent predictive accuracy.

Random Forest ($R^2 = 0.8413$) underperformed relative to Linear Regression. This is most likely because the relationship between features and waste is largely linear and

time-driven, which linear models handle more effectively than ensemble methods on small datasets of only 60 records.

## 4.4 Feature Importance




**(a)** Linear Regression coefficients. **(b)** Random Forest feature importances.

**Figure 5:** Feature importance analysis. Both models identify `Generation_Rate_kg_capita_day` and `Population` as primary drivers. Seasonal features (`Month`, `Is_Monsoon`, `Is_Summer`) also contribute significantly.

Figure 5a shows that `Generation_Rate_kg_capita_day` and `Population` carry the largest Linear Regression coefficients, confirming their role as primary waste drivers. Figure 5b confirms that Random Forest similarly ranks `Population` and `Waste_Per_Capita` as top predictors, with `Month` and `Quarter` capturing the strong temporal trend.

## 4.5 Actual vs. Predicted

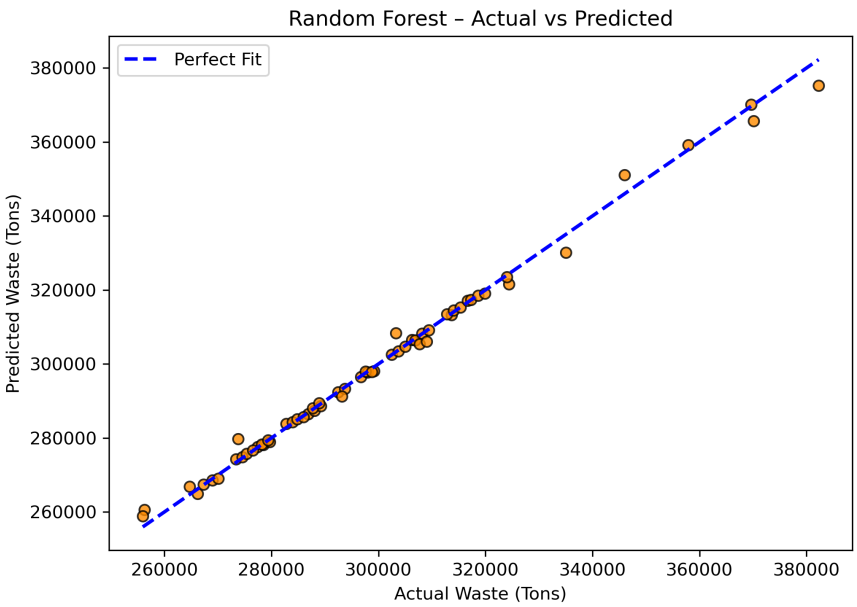
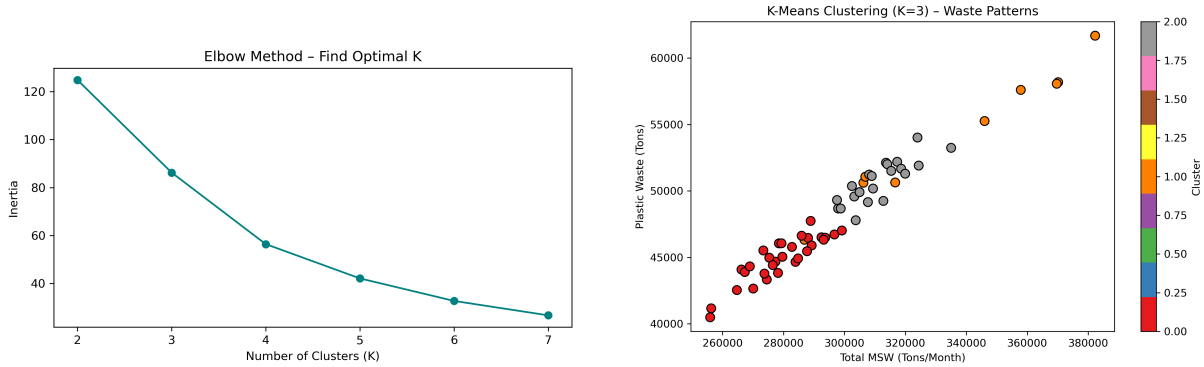



**Figure 6:** Random Forest: actual vs. predicted values. Most points cluster tightly around the perfect-fit diagonal at lower waste volumes; larger deviations appear at higher tonnage, reflecting the known limitation of Random Forest on extrapolation beyond the training range.

## 4.6 Clustering Results




**(a)** Elbow method: inertia vs. $K$.

**(b)** K-Means scatter plot ($K = 3$).

**Figure 7:** Clustering analysis. The elbow confirms $K = 3$ as optimal; the scatter plot reveals three distinct waste-generation regimes.

Figure 7a confirms that $K = 3$ is the optimal number of clusters, where the inertia curve shows a clear elbow; beyond $K = 3$, additional clusters provide diminishing inertia reduction.

Figure 7b reveals three distinct waste-generation patterns:

- **Cluster 0** (30 months): Moderate waste volumes — typical non-seasonal months with average population and generation rate.
- **Cluster 1** (9 months): High waste volumes with elevated plastic content — likely summer and event months with increased consumption.
- **Cluster 2** (21 months): Lower waste volumes — cooler, non-event months with below-average generation rates.

## 4.7 Discussion

Linear Regression outperformed Random Forest because total MSW generation in Karachi follows a strong, nearly linear upward trend driven by population growth and increasing per-capita waste generation. This deterministic temporal relationship is captured perfectly by a linear model but is harder for a tree-based ensemble to learn on only 60 data points.

The clustering analysis adds valuable insight beyond regression, revealing that Karachi's monthly waste does not follow a single uniform pattern. The three distinct clusters suggest different operational contexts for waste management — routine months, high-waste periods, and lower-waste periods — which can guide dynamic resource allocation for collection vehicles and disposal facilities.

## 5 Conclusion

A complete data mining pipeline was developed to analyse and predict Karachi's municipal solid waste generation, covering data loading, preprocessing, exploratory analysis, feature engineering, regression modelling, and clustering.

**Key outcomes:**

- No missing values were found in the dataset — data quality was high throughout.
- **Linear Regression** emerged as the best model with $R^2 = 0.9984$, MAE = 615.16 tons, confirming strong linear predictability of monthly MSW.
- **Random Forest** ($R^2 = 0.8413$) was less effective on this small, time-driven dataset.
- **Three distinct waste clusters** were identified using K-Means, revealing seasonal and event-driven patterns in Karachi's waste behaviour.
- **Population** and **Generation_Rate_kg_capita_day** were consistently the most important predictors across both models.

Future work could explore deep learning or LSTM-based time-series models with extended datasets to further improve long-term waste predictions and support Karachi's smart city waste management initiatives.